

Power and Efficiency & Equilibrium Review

The rate of doing work, the efficiency of a machine, and a full review of a body held in equilibrium by cables and springs.

Grade 11 · Egyptian Baccalaureate

22 questions

Step-by-step solutions

 $g = 9.8 \text{ m/s}^2$

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PHYSICS

The toolkit for this sheet

$$P = \Delta W / \Delta t$$

average power — unit : watt

$$P = F v \cos \theta$$

power of a force on a moving body

$$\eta = \text{useful output} / \text{total input}$$

efficiency, always < 100 %

$$W = m g h$$

work done in raising a load

$$\Sigma F_x = 0, \Sigma F_y = 0$$

equilibrium on each axis

$$T = W / (2 \sin \alpha)$$

two symmetric cables carrying a weight

Watch the difference. Work tells you *how much* energy was transferred; power tells you *how fast* it was transferred. Two machines can do exactly the same work and still have very different powers.

LESSON 1-6 · CLASSWORK 1

A

Power and efficiency

1 Power is a measure of:

MCQ

A the amount of work done only

B the rate of doing work or of transferring energy

C the magnitude of the force only

D the total displacement of the body

2 The mathematical expression used to calculate the average power from the work and the time is:

MCQ

A $P = W \times t$

B $P = \Delta W / \Delta t$

C $P = W + t$

D $P = W / t^2$

3 The SI unit of power is:

MCQ

A the joule (J)

B the watt (W)

C the newton (N)

D the kilogram (kg)

[EXPLAIN](#)

- 4 The figure shows two lifts A and B raising the **same** load through the **same** height, but in different times. **Explain which lift has the greater power**, and **which one does the greater useful work**. ($g = 9.8 \text{ m/s}^2$)



FIGURE 1

LESSON 1-6 · HOME ASSIGNMENT 1

B Using power and efficiency

- 5 When two lifts raise the same load through the same vertical height, but one of them finishes the lift in a shorter time, then this lift has:

[MCQ](#)

- ☐ A a greater useful work
- ☐ B a greater power
- ☐ C necessarily a greater efficiency
- ☐ D the same work and the same power

- 6 A machine has an efficiency of 40 %. If the total input energy supplied to it is 250 J, then the useful output energy equals:

[MCQ](#)

- ☐ A 100 J
- ☐ B 250 J
- ☐ C 62.5 J
- ☐ D 400 J

- 7 An electric water heater receives an electric power of 2400 W and produces a useful thermal power of 2040 W. Its efficiency equals:

[MCQ](#)

- ☐ A 85 %
- ☐ B 15 %
- ☐ C 117 %
- ☐ D 60 %

8

The figure shows a force F acting at an angle θ to the direction of motion of a body moving with a velocity v . **Explain what happens to the power produced by the force as θ gets closer to 90°** , compared with the case in which the force acts exactly along the direction of motion ($\theta = 0^\circ$).

EXPLAIN

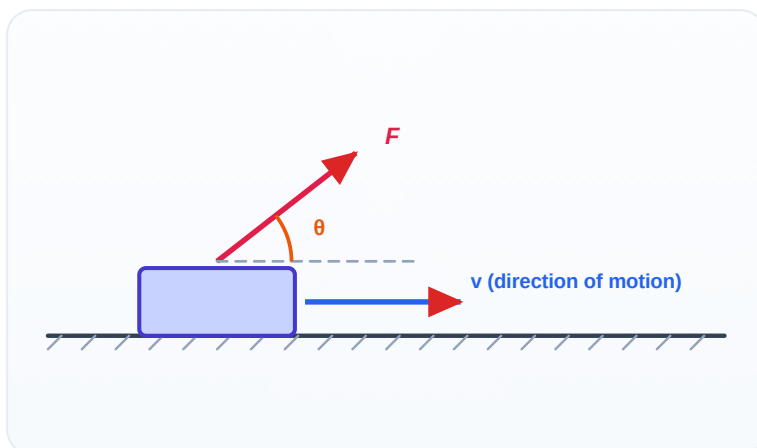


FIGURE 2

A body in equilibrium under several forces

- 9 A rigid body is in equilibrium under the action of three forces only: $F_1 = 10\text{ N}$ upwards and $F_2 = 6\text{ N}$ downwards. The third force F_3 needed to keep the body in equilibrium (on the same vertical line) must be:

MCQ

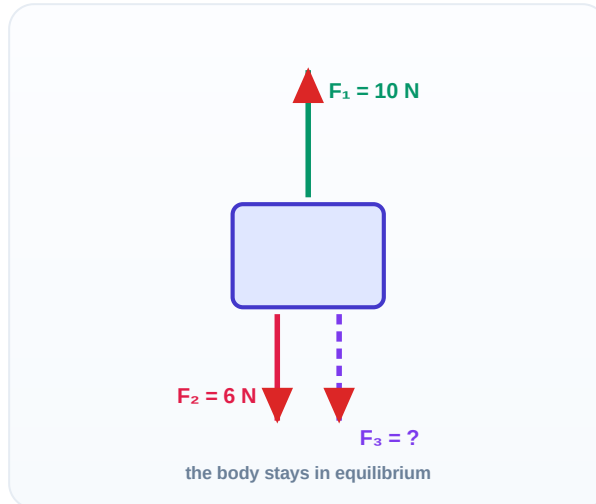


FIGURE 3

- A 4 N upwards
- B 4 N downwards
- C 16 N upwards
- D zero

- 10 A lamp of mass 3.0 kg hangs from a cable fixed to the ceiling together with a light horizontal rope fixed to a wall, and the cable makes an angle of 25° with the vertical. The weight of the lamp is 29.4 N , so the tension T in the cable is approximately:

MCQ

- A 26.8 N
- B 32.3 N
- C 12.4 N
- D 29.4 N

- 11 For a lamp hanging in equilibrium from two identical cables that make the **same** angle with the horizontal, the two horizontal components of the tensions:

MCQ

- A are equal in magnitude and opposite in direction, so they cancel each other
- B add together
- C have nothing to do with the equilibrium
- D each one equals the weight

- 12 If the angle that each cable makes with the horizontal is **decreased** (so that the two cables become closer to the horizontal) while the weight of the load stays constant, then the tension in each cable:

MCQ

- A increases
- B decreases
- C stays constant
- D becomes zero

- 13 If the angle that each cable makes with the horizontal is **increased** (so that the two cables become closer to the vertical) while the weight of the load stays constant, then the tension in each cable:

MCQ

- A increases
- B decreases
- C stays constant
- D becomes zero

- 14 A metal piece of mass 1.5 kg hangs at rest from two identical light springs, each making an angle of 40° with the vertical, the spring constant of each being 60 N/m. The weight of the piece is approximately: ($g = 9.8 \text{ m/s}^2$)

MCQ

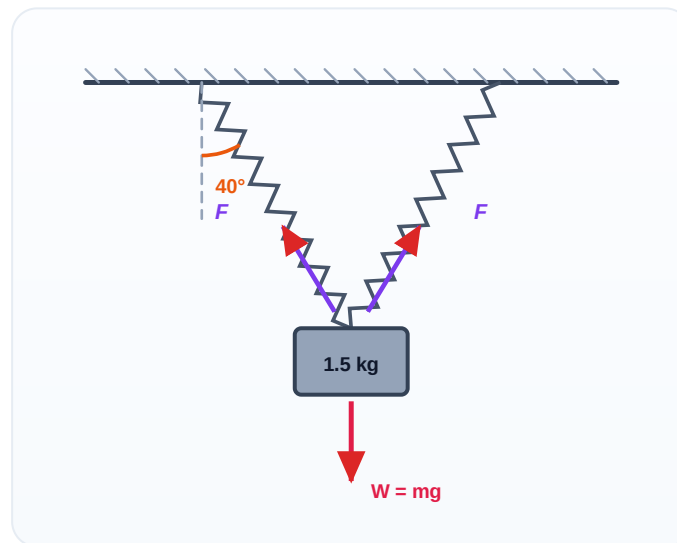


FIGURE 4

- A 1.5 N
- B 14.7 N
- C 6.0 N
- D 60 N

- 15 For the same metal piece of the previous question, calculate the force F in **each** spring. ($\cos 40^\circ \approx 0.77$)

PROBLEM

- 16 The figure shows a lamp of mass 4.0 kg hanging from a cable fixed to the ceiling and a light horizontal rope fixed to a wall. The tension measured in the horizontal rope is $P = 15$ N. Calculate the tension T in the cable.

PROBLEM

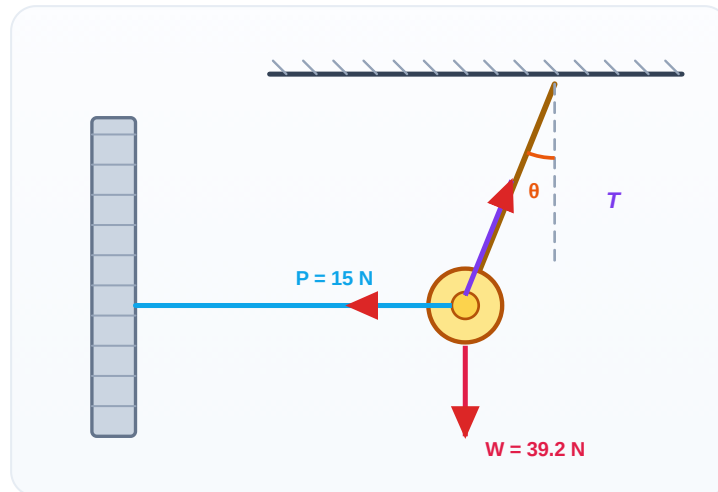


FIGURE 5

- 17 For the same lamp of the previous question, find the angle θ that the cable makes with the vertical.

PROBLEM

- 18 Using the figure, explain how the diagram shows that the sum of the two vertical components of the tensions equals the weight of the lamp, while the two horizontal components cancel each other.

EXPLAIN

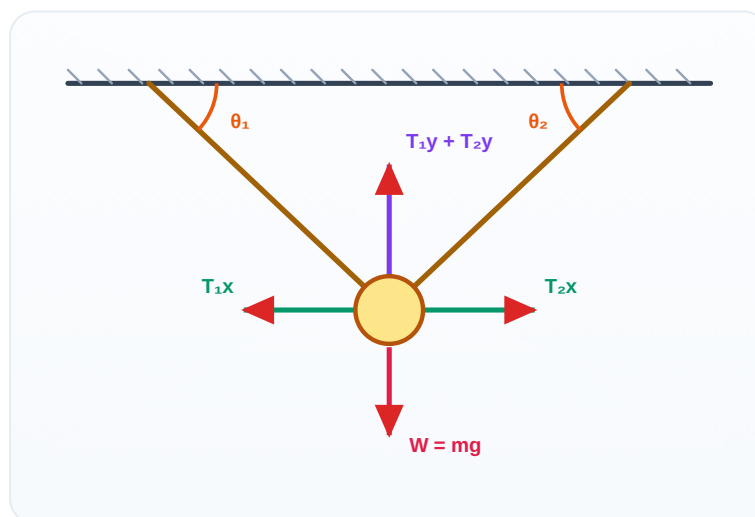


FIGURE 6

- 19 Compare the case of a lamp hanging from two identical cables making the same angle with the case of another lamp of the same weight hanging from two cables making different angles, in terms of whether the tensions in the two cables are equal or different.

COMPARE

- 20 The figure shows a shop sign of mass 10 kg hanging from two light cables attached to a horizontal beam. The first cable makes an angle of 35° with the horizontal and the second makes 65° with the horizontal. **Find the tension in each cable.** ($g = 9.8 \text{ m/s}^2$)

PROBLEM

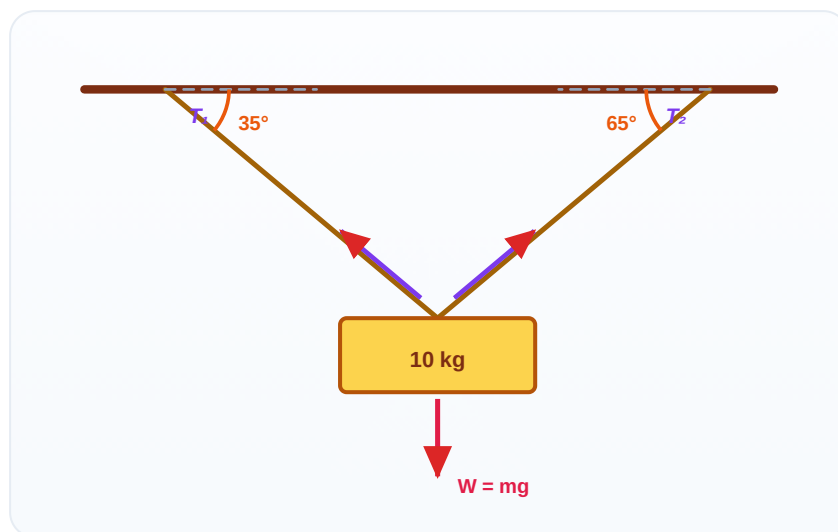


FIGURE 7

- 21 For a lamp of weight W hanging in equilibrium from two identical cables making the same angle with the horizontal, calculate the **sum of the two vertical components** of the tensions.

PROBLEM

- 22 The figure shows a book resting on a table. **What changes in this diagram** if another book of the same weight is placed on top of the first one? State clearly which of the two forces changes in value and which one stays the same, and identify the **source** of each force.

EXPLAIN

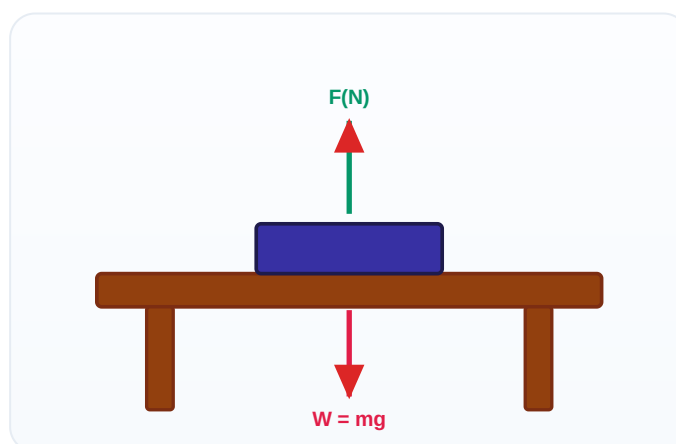


FIGURE 8



ANSWER KEY

Solutions — step by step

For every hanging body, write the two equilibrium equations separately: the horizontal one and the vertical one. Almost every question in this sheet falls out of those two lines.

Part A — Power and efficiency

1 what power measures

work tells how much energy was transferred;
power tells how fast it was transferred

B) the rate of doing work or of transferring energy

2 the relation

$$P = \Delta W / \Delta t \quad (\text{work done} \div \text{time taken})$$

B) $P = \Delta W / \Delta t$

3 the unit

$$1 \text{ watt} = 1 \text{ joule} / \text{second}$$

B) the watt (W)

4 the two lifts

the two lifts raise the same load through the same height, so the useful work is the same :

$$W = m g h = 600 \times 9.8 \times 9.0 = 52\,920 \text{ J} \approx 5.3 \times 10^4 \text{ J}$$

$$\text{lift A : } P = W / t = 52\,920 / 24 = 2205 \text{ W}$$

$$\text{lift B : } P = W / t = 52\,920 / 12 = 4410 \text{ W}$$

Power is not "more work", it is the **same** work delivered in less time. Lift B finishes in half the time, so its power is twice as large.

the same useful work; lift B has double the power

Part B — Home assignment 1

5 the faster lift

same W , smaller $t \rightarrow P = W / t$ is larger

B) a greater power

6 useful output from the efficiency

$$\eta = \text{useful output} / \text{total input}$$

$$\text{useful output} = \eta \times \text{input} = 0.40 \times 250 = 100 \text{ J}$$

A) 100 J

7 the efficiency of the heater

$$\eta = \text{useful power} / \text{input power} = 2040 / 2400 = 0.85 = 85 \%$$

An efficiency greater than 100 % is impossible, which rules out the 117 % option immediately.

A) 85 %

8 the power of an inclined force

$$P = F v \cos \theta$$

$$\theta = 0^\circ : \cos \theta = 1 \rightarrow P = F v \quad (\text{the largest possible power})$$

$$\theta \rightarrow 90^\circ : \cos \theta \rightarrow 0 \rightarrow P \rightarrow \text{zero}$$

Only the component of the force along the motion, $F \cos \theta$, transfers energy to the body; a force perpendicular to the motion does no work at all.

the power falls from its maximum $F v$ down to zero

Part C — Equilibrium: review and weekly assessment

9 the third force

$$\Sigma F(y) = 0$$

$$+ 10 - 6 + F_3 = 0 \rightarrow F_3 = -4 \text{ N}$$

the minus sign means: directed downwards

B) 4 N downwards

10 the tension in the cable

the rope is horizontal, so the vertical equation contains the cable only :

$$T \cos 25^\circ = W = 29.4 \text{ N}$$

$$T = 29.4 / 0.906 = 32.4 \text{ N}$$

B) 32.3 N

11 the horizontal components

$\Sigma F(x) = 0$, and the two cables are symmetric, so the two horizontal components have the same size and opposite directions

A) equal and opposite, so they cancel each other

12 cables closer to the horizontal

$$T = W / (2 \sin \alpha)$$

α decreases $\rightarrow \sin \alpha$ decreases $\rightarrow T$ increases

As the cables approach the horizontal, $\sin \alpha$ approaches zero and the tension grows without limit — this is why a washing line pulled dead straight snaps so easily.

A) increases

13 cables closer to the vertical

$$T = W / (2 \sin \alpha)$$

α increases $\rightarrow \sin \alpha$ increases $\rightarrow T$ decreases

B) decreases

14 the weight of the metal piece

$$W = m g = 1.5 \times 9.8 = 14.7 \text{ N}$$

The spring constant and the angle are not needed for the weight; they are needed for the next question.

B) 14.7 N

15 the force in each spring

the two springs are identical, so each carries the same force F .

each spring makes 40° with the **vertical**, so its vertical component is $F \cos 40^\circ$:

$$\Sigma F(y) = 0 \rightarrow 2 F \cos 40^\circ = W$$

$$2 F \times 0.77 = 14.7$$

$$F = 14.7 / 1.54 = 9.5 \text{ N}$$

F $\approx$ 9.5 N in each spring

16 the tension from the horizontal rope

the cable has to balance both the weight and the pull of the rope :

$$\text{vertical} : T \cos \theta = W = m g = 4.0 \times 9.8 = 39.2 \text{ N}$$

$$\text{horizontal} : T \sin \theta = P = 15 \text{ N}$$

$$T = \sqrt{W^2 + P^2} = \sqrt{39.2^2 + 15^2} = \sqrt{1761.6} = 42.0 \text{ N}$$

T = 42.0 N

17 the angle of the cable

$$\tan \theta = P / W = 15 / 39.2 = 0.3827$$

$$\theta = 20.9^\circ$$

$\theta \approx 20.9^\circ$ with the vertical

18 reading the component diagram

vertical axis : $\Sigma F(y) = 0 \rightarrow T_1 y + T_2 y - W = 0 \rightarrow T_1 y + T_2 y = W$

horizontal axis : $\Sigma F(x) = 0 \rightarrow T_1 x = T_2 x$ (opposite directions)

The weight has no horizontal component at all, so nothing is left for the horizontal components to balance except each other.

the vertical components carry the weight, the horizontal ones cancel

19 equal angles against different angles

equal angles : by symmetry the two tensions are equal , $T = W / (2 \sin \alpha)$

different angles : the tensions are **not** equal; the cable that is closer to the vertical (the larger angle with the horizontal) carries the larger tension

equal angles $\rightarrow$ equal tensions ; different angles $\rightarrow$ different tensions

20 the shop sign on two cables

$$W = m g = 10 \times 9.8 = 98 \text{ N}$$

horizontal : $T_1 \cos 35^\circ = T_2 \cos 65^\circ$

$$0.819 T_1 = 0.423 T_2 \rightarrow T_1 = 0.516 T_2$$

vertical : $T_1 \sin 35^\circ + T_2 \sin 65^\circ = 98$

$$0.574 (0.516 T_2) + 0.906 T_2 = 98$$

$$1.202 T_2 = 98 \rightarrow T_2 = 81.5 \text{ N}$$

$$T_1 = 0.516 \times 81.5 = 42.1 \text{ N}$$

The steeper cable (65°) carries almost twice the tension, because it has to supply the larger share of the vertical support.

$T_1 = 42.1 \text{ N}$, $T_2 = 81.5 \text{ N}$

21 the sum of the vertical components

$$\Sigma F(y) = 0 \rightarrow T_1 y + T_2 y = W = m g$$

the sum equals the weight of the lamp

22 a second book on the table

the table now has to support two books, so the normal force becomes :

$$F(N) = 2 m g \quad (\text{it doubles})$$

the weight of the first book is still $m g$, because it depends only on its own mass and on g .

The two forces have different **sources**: the weight is the gravitational pull of the Earth on that one book, while the normal force is a contact reaction from the table that adjusts itself to carry whatever rests on it.

$F(N)$ doubles, the weight of the book itself does not change